

Free Resolutions

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This is a note discussing the basic construction of Free Resolutions and the computation of free resolutions for various modules.

1 Basic Construction

Let R be a commutative ring with 1 and let M be a finitely generated module over R . If M is generated by $x_1, \dots, x_n$ elements with m relations f_i between the generators then recall that we can encode the presentation of M with an exact sequence

$$R^m \rightarrow R^n \rightarrow M \rightarrow 0$$

where $R^n \rightarrow M$ is given by $a_i \rightarrow x_i$ and where $R^m \rightarrow R^n$ is given by $b_i \rightarrow f_i(a_i)$. This is already pretty good. However, we can go further. Recalling that $R\text{-mod}$ is an abelian category gives us that $\ker(R^m \rightarrow R^n)$ is also an R -module. And so we can ask for a presentation of $\ker(R^n \rightarrow M)$. This gives us a further exact sequence

$$R^{m_2} \rightarrow R^{n_2} \rightarrow R^m \rightarrow R^n \rightarrow M \rightarrow 0.$$

Then, we are able to iterate this procedure until we no longer have any relations between the set of generators of the current iterate. This process may not terminate (**even when M is fg? we should find an example**). Repeating this procedure gives the *free resolution* of M

$$\dots \rightarrow R^{k_n} \rightarrow \dots R^{k_2} \rightarrow R^{k_1} \rightarrow R^{k_0} \rightarrow M \rightarrow 0.$$

This is called “Free” since every object in the chain (except the final one) is a free module.

Here’s another way to think about this iterative process, which may help with the computations.

M is finitely generated with n_1 generators and so we start with the exact sequence

$$R^{n_1} \rightarrow M \rightarrow 0.$$

Now we ask for the generators of $\ker(R^{n_1} \rightarrow M)$, say this has n_2 generators. We can then extend our exact sequence

$$R^{n_2} \rightarrow R^{n_1} \rightarrow M \rightarrow 0$$

Now, ask for the generators of $\ker(R^{n_2} \rightarrow R^{n_1})$,

$$R^{n_3} \rightarrow R^{n_2} \rightarrow R^{n_1} \rightarrow M \rightarrow 0.$$

Then continue this process indefinitely, or until $\ker(R^{n_{k+1}} \rightarrow R^k) = 0$ is the zero module. **Is it clear that this kernel module should be finitely generated?**

Below we will compute some examples of free resolutions of various modules. ¹

2 Examples

These examples are sorted into an order of increasing complexity. However, in all the following examples, R is a polynomial ring and M is a quotient of R . One can check these examples in Macaulay2 with the code **what's the use of verbatim**

Example 1. Let $R = k[x]$ and $M = R/(x^n)$ for some fixed n . Then notice that M is generated by $\bar{1} \in R/(x^n)$ as an R -module. Moreover, we only have a single relation $x^n \cdot \bar{1} = 0$. We can then capture this in the presentation

$$R^1 \xrightarrow{1 \mapsto x^n} R^1 \xrightarrow{1 \mapsto \bar{1}} M \rightarrow 0.$$

The kernel of $R^1 \rightarrow M$ is exactly the ideal (x^n) (as expected). Interpreting (x^n) as an R -module we find that it is a rank 1 free module (generated by x^n). That is a presentation for (x^n) as an R -module is

$$0 \rightarrow R^1 \xrightarrow{1 \mapsto x^n} (x^n).$$

Is the above sequence correct? Now since $\ker(R^1 \rightarrow (x^n)) = 0$ we can no longer iterate our procedure.

¹I am curious to note some basic theory here. I suppose it is clear that a fg module should always have a free resolution. Is it clear that a given module has a unique free resolution (I suppose the above procedure can only spit out one exact sequence, but if we define a free resolution to be an exact sequence whose tail is the presentation then is there only one such sequence?)

This leaves us with a (the?) free resolution

$$R^1 \xrightarrow{1 \mapsto x^n} R^1 \xrightarrow{1 \mapsto x^n} R^1 \xrightarrow{1 \mapsto \bar{1}} M \rightarrow 0.$$

The final map does not give us any additional information (what do I mean by this? It's clear that we should trim the sequence however) and so we actually call the following sequence the free resolution of $M = R/(x^n)$.

$$R^1 \xrightarrow{1 \mapsto x^n} R^1 \xrightarrow{1 \mapsto \bar{1}} M \rightarrow 0.$$

This example shows us that whenever our kernel module is free we are done.

Example 2. Let $R = k[x, y]$ and $M = R/(x^2, y^2)$ elements in M are of the form $f = a_0 + a_1x + a_2y + a_3xy$. As an R -module M is generated by $\bar{1} \in M$. We have the relations $x^2 = 0$ and $y^2 = 0$. We need to describe what it means for these relations to span all the kernel relations. A presentation for M is thus

$$R^2 \xrightarrow{\begin{bmatrix} x^2 & y^2 \end{bmatrix}} R^1 \xrightarrow{1 \mapsto \bar{1}} M \rightarrow 0$$

Now, the kernel module $\ker(R^1 \rightarrow M) = (x^2, y^2)$, interpreting the ideal as an R -module. This module is generated by two elements x^2, y^2 and we have the R -linear relation that $y^2(x^2) - x^2(y^2) = 0$. Thus our next iterated sequence is

$$R^1 \xrightarrow{\begin{bmatrix} y^2 \\ -x^2 \end{bmatrix}} R^2 \xrightarrow{\begin{bmatrix} x^2 & y^2 \end{bmatrix}} R^1 \xrightarrow{1 \mapsto \bar{1}} M \rightarrow 0.$$

Moreover the kernel module $\ker(R^2 \rightarrow (x^2, y^2)) = 0$ and so we are done. Put another way $\ker(R^1 \rightarrow R^2) = 0$. The above sequence is the free resolution for $M = R/(x^2, y^2)$.

Example 3. Let $R = k[x, y]$ and $M = R/(x, y)^2$. Note $(x, y)^2 = (x^2, xy, y^2)$. Now M is generated as an R -module by $1 \in M$ but we have the relations $x^2 = xy = y^2 = 0$. This gives the presentation

$$R^3 \xrightarrow{\begin{bmatrix} x^2 & -xy & y^2 \end{bmatrix}} R^1 \xrightarrow{1 \mapsto \bar{1}} M \rightarrow 0.$$

Now consider the kernel module $\ker(R^1 \rightarrow M) = (x^2, xy, y^2)$. Following the previous example, this has three generators x^2, xy, y^2 . We want to understand the kernel module $\ker(R^3 \rightarrow (x^2, xy, y^2))$. If $(a, b, c) \in R^3$ then this triple is in the kernel module when $ax^2 + bxy + cy^2 = 0$. Notice that the two vectors $(y, -x, 0)$ and $(0, y, -x)$ are both vectors in the kernel module. We claim that these are in fact an R -basis of $\ker(R^3 \rightarrow (x^2, xy, y^2))$. The argument we have is a direct computation and is a little tedious. For clarity here, we will put the details of this argument in the appendix.

Now, continuing the process, we want to understand $\ker(R^2 \rightarrow R^3)$ where $R^2 \rightarrow R^3$ is the map $(a, b) \mapsto (ay, -ax + by, -bx)$. That is, we want to understand for which $[a, b]^T \in R^2$ we have

$$\begin{bmatrix} ay \\ -ax + by \\ -bx \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

However, since R is a domain, we have that this is only satisfied by $a = b = 0$. Thus, $\ker(R^2 \rightarrow R^3) = 0$ and we are done. Our free resolution of $M = R/(x, y)^2$ is thus

$$R^3 \xrightarrow{\begin{bmatrix} y & 0 \\ -x & y \\ 0 & -x \end{bmatrix}} R^2 \xrightarrow{\begin{bmatrix} x^2 & xy & y^2 \end{bmatrix}} R^1 \xrightarrow{1 \mapsto \bar{1}} M \rightarrow 0.$$

I think we have written out the examples we wanted to discuss. We changed our understanding of how the free resolution is computed, and so we should go back through our examples and make sure that the verbiage is consistent. We also need to write the appendix with the calculation that we promised.